

A New Approach to Learn Spatio-Spectral Texture Representation with Randomized Networks: Application to Brazilian Plant Species Identification

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Abstract. Texture and color are fundamental visual descriptors, each complementing the other. Although many approaches have been developed for color-texture analysis, they often lack spectral analysis of the image and suffer from limited data availability for training in various problems. This paper introduces a new single-parameter texture representation, which integrates spatial and spectral analyses by combining the weights of the output layers of randomized autoencoders applied on both the same and adjacent image channels. As our approach is not end-to-end, we can extract individual representations for each image independently of the dataset size and without the need of fine-tuning. The rationale behind this approach is to learn meaningful spatial and spectral information of color-texture images through a simple neural network architecture. The proposed representation was evaluated using four benchmark datasets: Outex, USPtex, 1200Tex and MBT. We also verify the performance of the proposed representation on a practical and challenging task of Brazilian plant species identification. The experiments reveal that our method has a competitive classification accuracy in both scenarios when compared to the other methods, including various complex deep learning architectures. This shows an important contribution to the color-texture analysis and serves as a useful resource for other areas of computer vision and pattern recognition.

Keywords: Color-texture · Representation learning · Randomized neural network.

1 Introduction

Texture, a visual attribute known for its significant discriminatory properties, has been the subject of extensive study since the 1960s, in which it can be viewed as the spatial arrangement of intensity or visual patterns of the pixels within an image [15]. The texture is an object characteristic that allows them to be distinguished not only by the visual perception of the living beings, but also

by texture analysis and recognition approaches in digital images. They play a key role in various fields and problems, such as in botanic [6], material sciences [19], medicine [40, 12], agriculture [3], and others.

In that sense, although for a period of time only the use of texture was needed, currently computer vision and pattern recognition applications are increasingly filled with colored images, demanding more powerful color-texture analysis approaches. Thereby, since it has been shown that the use of color, a powerful visual cue, enhances the overall image description in pattern recognition tasks, many color-texture methods have been proposed [33, 15, 5, 28]. These works indicate that texture and color are fundamental visual elements within an image, functioning as foundational components across numerous applications and within various domains of computer vision [37].

Several approaches for color-texture analysis have been proposed in the literature. Most of these methods use the integrative approach which applies gray-level texture techniques on each color channel separately and combines these features to form the representation. Examples include integrative local binary pattern methods such as integrative CLBP [11], LETRIST [38], and LGONBP [36], with the latter two enhancing the noise robustness and rotation invariance, and complex network approaches such as integrative CNDT [4]. Additionally, deep convolutional neural networks (DCNNs) have been used for learning color-texture representations [39], although they still require large computational costs and still suffer from data availability. On the other hand, there are also pure color methods that ignore the spatial relationship of the pixels, such as color statistics methods. Hence, a widely used approach is to combine both methods for accounting texture and color attributes [9].

Further, over the past few years, it is noteworthy that many texture representation learning techniques, as presented in the literature [30, 31, 27, 29], use randomized neural networks due to their simplicity, low computational cost, and high accuracies, allowing them to be applied in many tasks, especially real-time ones. However, these color-texture analysis approaches lack incorporation of spectral analysis of the image, applying only a parallel methodology [15] for acquiring texture representation. Thus, they overlook the potential benefits of employing spectral analysis, which can provide deeper insights into essential aspects of the image.

This paper proposes a new single-parameter texture representation that integrates spatial and spectral analysis using randomized networks. First, we divide each image channel into stretched 3×3 patches that sweep over every pixel of the image to assemble the input feature matrix. Next, we employ randomized neural network-based modules to conduct spatial and spectral analysis, using the same input and adjacent image channel input feature matrix as input and output, respectively. Thus, the proposed descriptor is composed by learned weights of the output layer, which extract spatial and spectral texture key information within and between color channels. In summary, the major contributions of the proposed approach are as follows: (i) A simple, fast, and single-parameter approach based on randomized neural networks. (ii) A texture descriptor containing key

information about the relationship of the multispectral pixels obtained from spectral analysis. (iii) A texture representation demonstrating high performance in the practical application of identifying plant species utilizing images from foliar surfaces.

The paper is organized as follows. In Section 2, we describe the proposed texture representation. In Section 3 the experimental setup, parameter evaluation and the comparison of the results are shown, followed by the conclusion in Section 4.

2 Proposed Method

2.1 Randomized Neural Network

Randomized neural networks (RNNs) are one-pass artificial neural networks that were proposed in [34, 24, 25, 14]. In their simplest form, they consist of a single hidden layer with randomly generated weights following some probability distribution. The primary objective is to non-linearly project the input data into another dimensional space, thereby enhancing the likelihood that the data will become linearly separable, as stated in Cover’s theorem [10]. Ultimately, the learning is performed through the weights of the output layer, which can be calculated using the Moore-Penrose pseudoinverse [26].

Mathematically, consider the input matrix $X \in \mathbb{R}^{p \times N}$ composed of N samples of p -dimensional feature input vectors, and the output matrix $Y \in \mathbb{R}^{r \times N}$, comprising N samples of r -dimensional output vectors. To initiate the process, the hidden layer weight matrix $W \in \mathbb{R}^{Q \times (p+1)}$ is generated, with Q denoting the number of hidden neurons and p representing the number of attributes in each feature input vector.

Additionally, consider the row matrix $\mathbb{1}_N$ with N columns, each entry set to one. This allows the construction of the matrix $X' = [-\mathbb{1}_N^T \ X^T]^T$, where the first row corresponds to the bias. Subsequently, the forward pass of the hidden layer is computed as $U = \phi(WX')$, where ϕ denotes a sigmoid function, and in which the columns of matrix U represent the projected feature input vectors.

Further, defining $Z = [-\mathbb{1}_N^T \ U^T]^T$ as the matrix U augmented with the bias, we determine the learned weights for the output layer using:

$$M = YZ^T (ZZ^T)^{-1}, \quad (1)$$

where $Z^T(ZZ^T)^{-1}$ represents the Moore-Penrose pseudoinverse [20, 26]. It is worth noting that the matrix ZZ^T may become singular or nearly singular, leading to unstable outcomes. To mitigate such scenarios, Tikhonov’s regularization [1, 7] can be applied by modifying the formula to $M = YZ^T (ZZ^T + \lambda I)^{-1}$, where $\lambda > 0$, and I denotes the identity matrix.

Lastly, RNNs can function as a randomized autoencoder (RAE) by treating the input feature matrix X as the target output Y (i.e., $Y = X$). In this scenario, the model consists of a random encoder and a decoder based on least squares, capable of mapping the input data. Consequently, the weight matrix M serves as

the transformation for projecting the randomly generated Q -dimensional space back into the original input data.

2.2 Learning color-texture representations

In this paper, we propose to use as texture representation the weights of the output layer of randomized networks and autoencoders trained with information within and between image channels to conduct a spatio-spectral analysis.

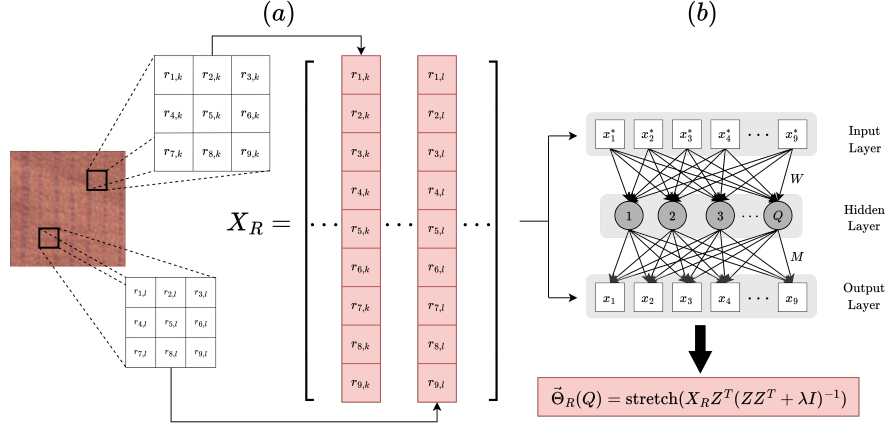
For this, let I be a 3-channel image, with I_C denoting a channel of I , where $C \in \{R, G, B\}$. We define X_C as the matrix built by assembling stretched 3×3 windows that traverse the entirety of channel C , i.e., the window is centered in each pixel of the image and overlaps each other. The process to build the input feature matrix X_R for the red channel is illustrated in Figure 1(a). Note that the X matrices are constructed from individual images, making the several randomized networks to be trained for each image in our approach. Further, the weight matrix W for the hidden layer is generated using a Linear Congruent Generator (LCG) with the equation $V(n+1) = (aV(n) + b) \bmod c$, where V has a length of $L = Q \cdot (p+1)$, and parameters $V(0) = L+1, a = L+2, b = L+3$, and $c = L^2$. Hence, the matrix W is formed by partitioning the vector V into Q segments, each of length $p+1$. Furthermore, all values in matrix W undergo normalization through z -scoring. Ultimately, Tikhonov's regularization was used with $\lambda = 10^{-3}$ as commonly employed [31, 28, 29] to prevent singular matrices.

Therefore, we propose two learning modules to construct two texture representations:

- **Spatial Representation** $\vec{\Theta}_C(Q)$: This module is responsible for the spatial representation, which is built by stretching the weights of the output layer of a randomized autoencoder, using a z -scored version of X_C as input matrix and the X_C itself as output matrix, as shown in Figure 1, conducting a spatial analysis. The spatial analysis enables the randomized network to learn the arrangement of the pixels in local neighborhoods, leading the trained weights of the output layer to capture essential details about the spatial relationship between the pixels.
- **Spectral Representation** $\vec{\Omega}_{C_1, C_2}(Q)$: This representation is obtained by stretching the weights of the output layer of a randomized network, using a z -scored version of X_{C_1} as input feature matrix and X_{C_2} itself as output feature matrix, as shown in Figure 2, conducting a spectral analysis. C_1 and C_2 are two adjacent channels such as (R, G) , (G, B) and (G, R) . The spectral analysis provides the means for the randomized network to learn to predict the pixels of one channel based on the previous one, thus causing the trained weights of the output layer to contain key information about the spectral relationship between two channels.

Therefore, with the two constructed signatures, $\vec{\Theta}_C(Q)$ and $\vec{\Omega}_{C_1, C_2}(Q)$, which conduct spatial and spectral analyses, respectively, we combine both to create our proposed spatio-spectral texture representation using all channels:

Fig. 1. Illustration of the proposed approach for conducting a spatial analysis. (a) Construction of X_R from assembling of stretched 3×3 windows of the red channel. (b) spatial representation $\vec{\Theta}_R(Q)$ from stretching the output layer weight matrix.



$$\vec{\Psi}(Q) = [\vec{\Theta}_R(Q), \vec{\Theta}_G(Q), \vec{\Theta}_B(Q), \vec{\Omega}_{R,G}(Q), \vec{\Omega}_{G,B}(Q), \vec{\Omega}_{B,R}(Q)] \quad (2)$$

To conclude, the main steps of the proposed color-texture descriptor are summarized in a high-level module diagram in Figure 3. The spatial and spectral modules have been detailed previously, and are characterized by the Figure 1 and 2. Each of these modules produces the spatial and the spectral representation, which will be concatenated later to form the proposed color-texture descriptor.

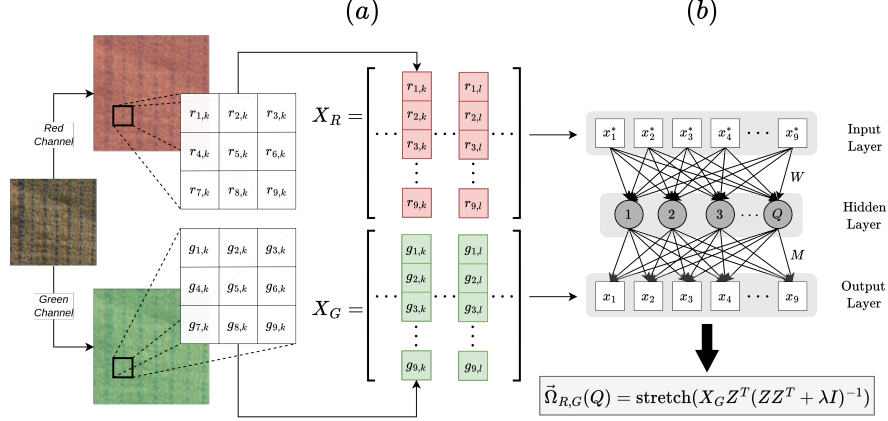
3 Results and Discussion

3.1 Experimental setup

The proposed approach described in Section 2 was applied in images of four color-texture datasets. The structure of the datasets are described as follows:

- **Outex** [21]: This dataset contains 1360 images divided into 68 classes with 20 images per class. The images in this dataset has size of 128x128 pixels.
- **USPtex** [5]: This dataset, from University of São Paulo, has 2292 images divided into 191 classes of natural textures, each having 12 images of 128x128 pixels.
- **MBT** [2]: The Multi-Band Texture dataset is composed by images obtained from the combination of intraband and interband spatial variations. It contains 2464 images divided into 154 classes, each having 16 images of 160x160 pixels.

Fig. 2. Illustration of the proposed approach for conducting a spectral analysis. (a) Construction of X_R and X_G from assembling of stretched 3×3 windows of the red channel. (b) spectral representation $\tilde{\Omega}_{R,G}(Q)$ obtained from stretching the output layer weight matrix.



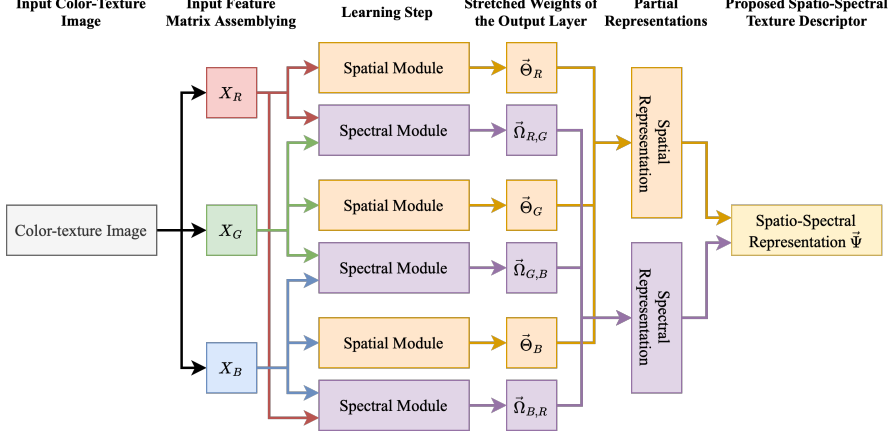
Finally, the texture representation is computed for each image, followed by a supervised classification using Linear Discriminant Analysis (LDA). Additionally, the performance of the classification is then evaluated using the leave-one-out cross-validation method.

3.2 Method investigation and parameter analysis

The proposed approach in Section 2 has the Q value as the unique parameter to calibrate. Therefore, we investigate the impact of Q values in our approach by computing the accuracy for various values of Q , as presented in Table 1.

The results show that the success rates tend to increase as Q increases. However, it is noteworthy that large feature vectors do not necessarily guarantee higher performances, as success rates tend to stabilize after a certain point. For instance, the representation $\tilde{\Psi}(13)$ shows the highest average accuracy without being the largest representation, indicating that larger representations do not necessarily improve performance. This is evident from the lower average accuracy obtained by the representation $\tilde{\Psi}(15)$. Furthermore, the representation $\tilde{\Psi}(03)$, comprising only 216 features, provides a good average accuracy with fewer features than $\tilde{\Psi}(09)$, $\tilde{\Psi}(13)$ and $\tilde{\Psi}(15)$, indicating a representation with a good trade-off between number of features and performance.

Moreover, although we do not use the reconstruction error as guidance for obtaining higher accuracies, in Figure 4 and Figure 5, we provided a visualization of the reconstructed images using the spatial $\tilde{\Theta}_C(Q)$ and spectral $\tilde{\Omega}_{C_1,C_2}(Q)$ representations, respectively. As can be seen, the reconstructed images are very similar to the original ones. This suggests that the two proposed modules are capable of learning the texture properties, with the learned output weights (decoder)

Fig. 3. A high-level diagram describing the proposed approach.**Table 1.** Accuracies of the feature vector $\tilde{\Psi}(Q)$ for various values of Q .

Q	No of Features	Outex	USPtex	MBT	Avg.
01	108	93.1	98.3	97.7	96.4
03	216	96.0	98.7	98.9	97.9
05	324	95.9	99.1	99.5	98.2
07	432	95.7	98.9	99.6	98.1
09	540	96.2	99.2	99.6	98.3
13	756	96.8	99.1	99.8	98.6
15	864	94.9	99.1	99.7	97.9

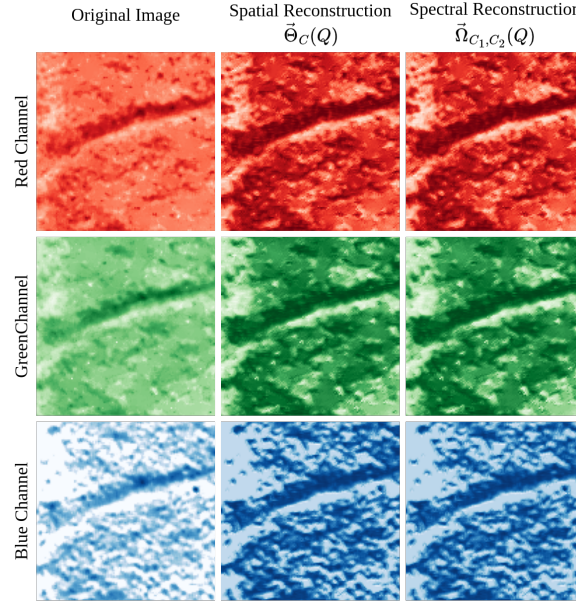
serving as valuable features to represent the images. Therefore, the capability of spatial and spectral representations (learned weights) to reconstruct the original image with good quality, as illustrated in the figures, provides clear evidence that the proposed descriptor captures rich information about the image. This highlights its potential as a highly discriminative and robust descriptor, capable of effectively distinguishing between different textures and colors in images.

3.3 Comparison and discussions

In order to evaluate the effectiveness of the proposed approach, we compared our best representation $\tilde{\Psi}(13)$ with other methods presented in the literature. Consistency was maintained across all methods with the use of the same experimental setup, employing the Linear Discriminant Analysis (LDA) classifier and using a leave-one-out cross-validation strategy. The results are shown in Table 2.

To begin with, our analysis compares the first section of methods which goes from Opponent-Gabor to SSR². In the Outex dataset, experimental results indicate that our proposed approach outperforms all compared methods in the

Fig. 4. Examples of the original images channels and the reconstructed images channels using the spatial $\tilde{\Theta}_C(Q)$ and spectral $\tilde{\Omega}_{C_1, C_2}(Q)$ representations, respectively, obtained from a sample of the 1200Tex dataset.

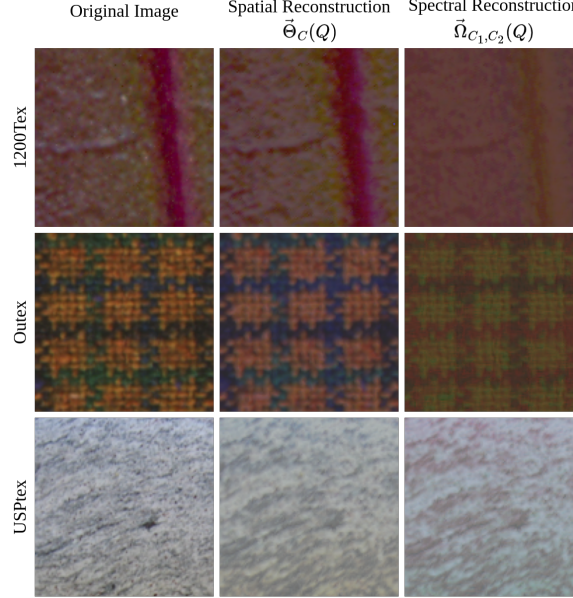


section, except for SSR² in which both obtain 96.8%. In the USPtex dataset, our method achieves the third-best accuracy, being just 0.6% of difference from the highest accuracy obtained by SSN (φ_{WB}^6). For the MBT dataset, our approach stands out, nearly reaching 100% accuracy and exceeding the accuracy of the second-best method by 1.2%. Further, it is noteworthy that our approach surpasses existing literature methods in the field of plant species recognition, a real, complex, and demanding problem presented in Section 3.4.

In relation to deep learning models, we compared results against various deep convolutional neural network architectures (DCNNs), including AlexNet [18], VGG16 and VGG19 [35], ResNet50 and ResNet101 [13], and InceptionV3 [39]. These results were obtained from [33], where these DCNNs were initially pre-trained on the ImageNet dataset, and then their pre-trained layers were used as a feature extraction method. Due to the fixed input size of each DCNN, images from each dataset were resized and inputted, generating the feature vectors by applying the Global Average Pooling (GAP) to the convolutional layers. For a comprehensive description of this methodology, refer to [33].

The proposed method demonstrated higher accuracy than DCNN architectures in both Outex and MBT datasets. However, for the USPtex dataset, the ResNet50 model slightly outperformed our proposed approach. In the MBT dataset, our proposed method outperforms all the DCNNs architectures. Particularly, the AlexNet, which achieved the highest accuracy among DCNN models

Fig. 5. Examples of the original images and their reconstruction using the spatial $\vec{\Theta}_C(Q)$ and spectral $\vec{\Omega}_{C_1,C_2}(Q)$ representations are along the columns. Each row contains a sample from a different dataset.



in the MBT dataset with 97.8%, was surpassed by our method which obtained 99.8%. The lower performance of these DCNNs on the MBT dataset can be attributed to the absence of spectral analysis, an important component for MBT. This significance arises from the structure of the dataset, where each image channel is composed of different textural images, emphasizing the importance of spectral analysis. In the Outex dataset, our proposed method surpasses by over 5% when compared to the various DCNNs architectures, showing the inefficiency of these models due to the varying illumination conditions present in the Outex dataset. For instance, the ResNet50, which represents the highest accuracy among the DCNNs models, obtained 91.5% while our method obtained 96.8%, showing a significant improvement.

Considering the results obtained from methods based on randomized neural networks, namely RNN-RGB, SSR¹, and SSR², our proposed approach outperforms RNN-RGB and SSR² across all datasets. Furthermore, it surpasses the SSR¹ in the Outex and MBT datasets, though SSR¹ maintains a slight edge with 0.2% higher accuracy in the USPtex dataset. However, it is particularly noteworthy that our approach enhances the accuracy by 1.2% in the MBT dataset. Therefore, this comparative analysis illustrates the advantage of integrating features extracted from both the same and adjacent image channels, thereby enhancing the overall accuracy, and evidencing the effectiveness of a spatio-spectral analysis of images.

Table 2. Comparison of accuracies of different literature methods on 3 color-texture datasets. The results of the deep convolutional networks were obtained from [33]. Unavailable results of some methods are represented by empty cells.

Method	USPTex	Outex	MBT
Opponent-Gabor [16]	99.1	93.5	97.6
LPQ integrative [23]	90.4	80.1	95.7
CLBP integrative [11]	97.4	89.6	98.2
CNTD integrative [4]	97.9	92.3	98.5
LETRIST (gray-level) [38]	92.4	82.8	79.1
LGONBP (gray-level) [36]	83.3	83.5	74.2
Sá Junior et al. [17]	96.9	91.5	—
MCND [32]	99.0	95.4	97.0
RNN-RGB [31]	98.4	94.8	—
SSN (φ_{WB}^6) [33]	99.7	96.6	98.6
SSR ¹ [28]	99.3	96.7	98.2
SSR ² [28]	99.0	96.8	98.0
AlexNet [18]	99.6	91.4	97.8
VGG16 [35]	99.5	91.1	97.2
VGG19 [35]	99.5	90.7	96.3
InceptionV3 [39]	99.5	89.5	94.4
ResNet50 [13]	99.7	91.5	94.9
ResNet101 [13]	99.5	91.3	94.6
Proposed Method			
$\vec{\Psi}(13)$	99.1	96.8	99.8

3.4 Brazilian plant species identification

To assess the practical application effectiveness of the proposed approach, we applied our method to plant species recognition. This poses challenges owing to the diverse spatial arrangements and color patterns found within a single plant leaf but also due to the resemblance in traits across different species [8]. Further, the plant species dataset [8], named 1200Tex, has 400 images divided into 20 classes, each having 20 images. However, for a leaf texture analysis, each image was cropped in three non-overlapping windows of 128x128 pixels, resulting in 1200 images. Figure 6 shows samples from different classes of the dataset. The experimental setup was the same one used in Section 3.1.

The results of the proposed approach $\vec{\Psi}(03)$ and other literature methods are shown in Table 3. The results show that our approach achieves the highest accuracy in recognizing plant species while maintaining the second-smallest texture representation size, comprising only 216 features. This shows an improvement over the second-best accuracy obtained by SSR¹, comprising 630 features, which uses a combination of graphs and randomized neural networks. Further, we also compare our representation with deep convolutional neural networks, such as AlexNet (256) [18], VGG16 (512) [35], VGG19 (512) [35] and ResNet101 (512) [13], in which the number in parentheses is the feature vector size, obtained by following the same feature extraction setup in [33]. Our proposed descriptor out-

Table 3. Comparison of accuracies of different literature methods in Brazilian plant species identification.

Method	Accuracy (%)
LPQ [23]	73.00
LBP [22]	72.00
CNTD [4]	83.33
$\vec{\Psi}(06, 08)_{04,14,19}$ [41]	87.00
SSN [33]	91.83
SSR ¹ [28]	94.08
SSR ² [28]	93.67
AlexNet [18]	84.92
VGG16 [35]	87.00
VGG19 [35]	83.92
ResNet101 [13]	93.42
Proposed Method	
$\vec{\Psi}(03)$	94.50

performed all compared DCNNs with an increase of 1.08% over the second-best DCNN accuracy, and with a lower feature vector size of 216 attributes, when compared to 512 features of ResNet101.

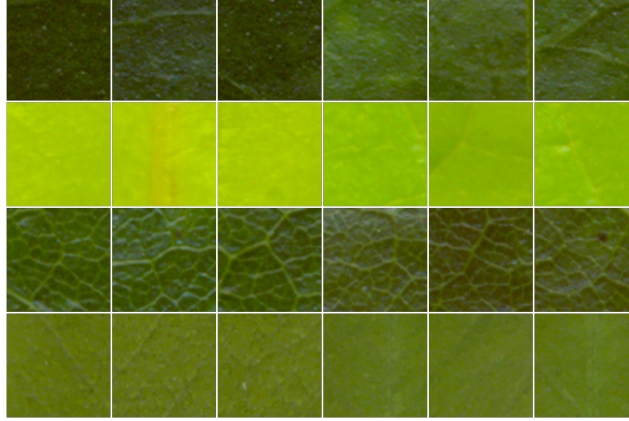
Finally, the SSR¹ and SSR² not only have higher feature vector size than $\vec{\Psi}(03)$ but also represent a multi-parameter method, contributing to the potential for time-consuming adaptation to newer situations due to the numerous possible combinations. On the other hand, our proposed approach has a smaller feature vector size, achieved the highest accuracy in the plant species recognition task and, primarily, it is a single-parameter method. These factors evidence the simplicity and effectiveness of our texture representation, indicating it to be a promising approach for computer vision and pattern recognition applications, such as plant species recognition.

4 Conclusion

In this paper, we proposed a new method for color-texture analysis which introduces a single-parameter texture representation that conducts a spatio-spectral analysis from the image color channels. In each image six randomized networks (or autoencoders) are applied, the first three conduct spatial analysis in each color channel, and the last three perform spectral analysis among each combination of two color channels.

The proposed method yields a texture representation with spatio-spectral properties, providing a highly discriminative color-texture representation. This is substantiated by the results obtained from benchmark datasets, where our method outperformed various texture literature methods. Furthermore, the proposed approach also outperformed other literature methods in a practical and challenging application: Brazilian plant species identification. In this context,

Fig. 6. Examples of samples from the 1200Tex dataset. This figure shows four different class with each row corresponding to six samples of the same class.



this paper demonstrates that the utilization of randomized network and autoencoder modules in color images holds significant potential for color-texture analysis and various computer vision tasks. As future work, we believe that our method can be adapted to other types of multi-channel images (e.g., hyperspectral) and other practical applications.

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References

1. On the solution of ill-posed problems and the method of regularization. On the solution of ill-posed problems and the method of regularization **151**, 501–504 (1963)
2. Abdelmounaime, S., Dong-Chen, H.: New brodatz-based image databases for grayscale color and multiband texture analysis. *ISRN Machine Vision* **2013**, 876386 (Feb 2013)
3. Babalola, E.O., Asad, M.H., Bais, A.: Soil surface texture classification using rgb images acquired under uncontrolled field conditions. *IEEE Access* **11** (2023)
4. Backes, A.R., Casanova, D., Bruno, O.M.: Texture analysis and classification: A complex network-based approach. *Information Sciences* **219**, 168–180 (2013)
5. Backes, A.R., Casanova, D., Bruno, O.M.: Color texture analysis based on fractal descriptors. *Pattern Recognition* **45**(5), 1984–1992 (2012)

6. Boudra, S., Yahiaoui, I., Behloul, A.: Tree trunk texture classification using multi-scale statistical macro binary patterns and cnn. *Applied Soft Computing* **118**, 108473 (2022)
7. Calvetti, D., Morigi, S., Reichel, L., Sgallari, F.: Tikhonov regularization and the l-curve for large discrete ill-posed problems. *Journal of Computational and Applied Mathematics* **123**(1), 423–446 (2000)
8. Casanova, D., de Mesquita Sá Junior, J.J., Bruno, O.M.: Plant leaf identification using gabor wavelets. *International Journal of Imaging Systems and Technology* **19**(3), 236–243 (2009)
9. Cernadas, E., Fernandez-Delgado, M., González-Rufino, E., Carrión, P.: Influence of normalization and color space to color texture classification. *Pattern Recognition* **61**, 120–138 (2017)
10. Cover, T.M.: Geometrical and statistical properties of systems of linear inequalities with applications in pattern recognition. *IEEE Transactions on Electronic Computers* **EC-14**(3), 326–334 (1965)
11. Guo, Z., Zhang, L., Zhang, D.: A completed modeling of local binary pattern operator for texture classification. *IEEE Transactions on Image Processing* **19**(6), 1657–1663 (2010)
12. Gómez Flores, W., de Albuquerque Pereira, W.C., Infantosi, A.F.C.: Improving classification performance of breast lesions on ultrasonography. *Pattern Recognition* **48**(4), 1125–1136 (2015)
13. He, K., Zhang, X., Ren, S., Sun, J.: Deep residual learning for image recognition. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. pp. 770–778 (2016)
14. Huang, G.B., Zhu, Q.Y., Siew, C.K.: Extreme learning machine: Theory and applications. *Neurocomputing* **70**(1), 489–501 (2006)
15. Humeau-Heurtier, A.: Color texture analysis: A survey. *IEEE Access* **10**, 107993–108003 (2022)
16. Jain, A., Healey, G.: A multiscale representation including opponent-color features for texture recognition. *IEEE Trans. Image Processing* **7**(1), 124–128 (1998)
17. Junior, J.J.d.M.S., Cortez, P.C., Backes, A.R.: Color texture classification using shortest paths in graphs. *IEEE Transactions on Image Processing* **23**(9), 3751–3761 (2014)
18. Krizhevsky, A., Sutskever, I., Hinton, G.E.: ImageNet Classification with Deep Convolutional Neural Networks. *Advances In Neural Information Processing Systems* pp. 1–9 (2012)
19. Liu, H., Fang, J., Xu, X., Sun, F.: Surface material recognition using active multi-modal extreme learning machine. *Cognitive Computation* **10**(6), 937–950 (Dec 2018)
20. Moore, E.H.: On the reciprocal of the general algebraic matrix. *Bulletin of American Mathematical Society* pp. 394–395 (1920)
21. Ojala, T., Mäenpää, T., Pietikäinen, M., Viertola, J., Kyllönen, J., Huovinen, S.: Outex - new framework for empirical evaluation of texture analysis algorithms. Object recognition supported by user interaction for service robots **1**, 701–706 vol.1 (2002)
22. Ojala, T., Pietikäinen, M., Maenpaa, T.: Multiresolution gray-scale and rotation invariant texture classification with local binary patterns. *IEEE Transactions on pattern analysis and machine intelligence* **24**(7), 971–987 (2002)
23. Ojansivu, V., Heikkilä, J.: Blur insensitive texture classification using local phase quantization (2008)

24. Pao, Y.H., Takefuji, Y.: Functional-link net computing: theory, system architecture, and functionalities. *Computer* **25**(5), 76–79 (1992)
25. Pao, Y.H., Park, G.H., Sobajic, D.J.: Learning and generalization characteristics of the random vector functional-link net. *Neurocomputing* **6**(2), 163–180 (1994)
26. Penrose, R.: A generalized inverse for matrices. *Mathematical Proceedings of the Cambridge Philosophical Society* **51**(3), 406–413 (1955)
27. Ribas, L.C., Junior, J.J.d.M.S., Scabini, L.F., Bruno, O.M.: Fusion of complex networks and randomized neural networks for texture analysis. *Pattern Recognition* **103**, 107189 (2020)
28. Ribas, L.C., Scabini, L.F., Condori, R.H., Bruno, O.M.: Color-texture classification based on spatio-spectral complex network representations. *Physica A: Statistical Mechanics and its Applications* p. 129518 (2024)
29. Ribas, L.C., Scabini, L.F., de Mesquita Sá Junior, J.J., Bruno, O.M.: Local complex features learned by randomized neural networks for texture analysis. *Pattern Analysis and Applications* **27**(1), 1–12 (2024)
30. Sá Junior, J.J.d.M., Backes, A.R.: Elm based signature for texture classification. *Pattern Recognition* **51**, 395–401 (2016)
31. Sá Junior, J.J.d.M., Backes, A.R., Bruno, O.M.: Randomized neural network based signature for color texture classification. *Multidimensional Systems and Signal Processing* **30**(3), 1171–1186 (2019)
32. Scabini, L.F., Condori, R.H., Gonçalves, W.N., Bruno, O.M.: Multilayer complex network descriptors for color-texture characterization. *Information Sciences* **491**, 30–47 (2019)
33. Scabini, L.F., Ribas, L.C., Bruno, O.M.: Spatio-spectral networks for color-texture analysis. *Information Sciences* **515**, 64–79 (2020)
34. Schmidt, W., Kraaijveld, M., Duin, R.: Feedforward neural networks with random weights. In: *Proceedings., 11th IAPR International Conference on Pattern Recognition. Vol.II. Conference B: Pattern Recognition Methodology and Systems.* pp. 1–4 (1992)
35. Simonyan, K., Zisserman, A.: Very deep convolutional networks for large-scale image recognition. *CoRR* **abs/1409.1556** (2014)
36. Song, T., Feng, J., Luo, L., Gao, C., Li, H.: Robust texture description using local grouped order pattern and non-local binary pattern. *IEEE Transactions on Circuits and Systems for Video Technology* **31**(1), 189–202 (2020)
37. Song, T., Feng, J., Wang, Y., Gao, C.: Color texture description based on holistic and hierarchical order-encoding patterns. In: *2020 25th International Conference on Pattern Recognition (ICPR).* pp. 1306–1312 (2021)
38. Song, T., Li, H., Meng, F., Wu, Q., Cai, J.: Letrist: Locally encoded transform feature histogram for rotation-invariant texture classification. *IEEE Transactions on circuits and systems for video technology* **28**(7), 1565–1579 (2017)
39. Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., Wojna, Z.: Rethinking the inception architecture for computer vision. In: *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (June 2016)
40. Xu, R., Cong, Z., Ye, X., Hirano, Y., Kido, S., Gyobu, T., Kawata, Y., Honda, O., Tomiyama, N.: Pulmonary textures classification via a Multi-Scale attention network. *IEEE J Biomed Health Inform* **24**(7), 2041–2052 (Nov 2019)
41. Zielinski, K.M.C., Ribas, L.C., Scabini, L.F.S., Bruno, O.M.: Complex texture features learned by applying randomized neural network on graphs. In: *2022 Eleventh International Conference on Image Processing Theory, Tools and Applications (IPTA).* pp. 1–6 (2022)